

# Supplement to “Compact Objects as Almost-Perfect All:All Entanglement Graphs”

Methods, honesty ledger, N-scaling, and module map (v5.0)

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## S1 Parameter audit and honesty ledger

Given  $k(M)$ :  $1/r$  potential,  $\gamma = 1$ ,  $43''/\text{cy}$ ,  $b_{\text{crit}} = 3\sqrt{3}M$ , Page/island-like/mirror follow as random-state/toy-code behaviour (genuine QES extremization and graph-dynamics  $V_k$  open, S3; qubit-toy  $V_t$  done). Not derived: mass–radius scaling itself,  $\beta(N)$  from geometry,  $w$  from the graph Laplacian (Damour–Schäfer from wiring), NICER  $M$ – $R$ – $\Lambda$  from routing stiffness, Kerr  $M_2/g_{t\phi}$ /ISCO/QNM from the graph.

Derived in-repo (new in v4.x):  $p$  from exact Ollivier–Ricci (not hard-coded),  $M(c_1, c_2)$  inversion,  $M_{\text{ej}}(M_{\text{tot}})$  scaling, peak-time ordering, band mags, O5 yield arithmetic,  $c$  from  $\ln 2$  scattering, UV pop as graph disconnection,  $p$  at  $N = 4000/8000/16000$  with turnover measured,  $d_{\text{eff}}$  bracket + convergence + shell/tense-plug pins (v0.6, S11).

Ruled out on the record: broad Planck-remnant dark matter (survives only in a  $\sim 0.4$ -dex EMD window at  $\sim 4 \times 10^5$  g); isotropic  $b_{\text{crit}} = 8M$ ; naive  $\gamma = 2$  strain; linear LIV; flat  $p = 0.49$  ( $-10\sigma$ ); bare shortest-path  $d(i, j)$  for tense regions (inverted vs GR); resistance/communicability distances (S11).

## S2 Gravity methods (AS, AT, AU, BB, BH, BI, BJ, BV)

**AS entropic Newton.** Leg screens + equipartition + Bekenstein bound give  $F = M_1 M_2 / r^2$  exact (slope  $-2$  to  $10^{-9}$ ); leapfrog orbits close,  $T^2 \propto r^3$ . Flux  $\propto 1/r^2$  distinguished from force (temperature corrects). Fig. 43.

**AT redshift.** GPS  $+5.29 \times 10^{-10}$  and Pound–Rebka  $2.55 \times 10^{-15}$  derived with no new parameters. Congestion fronts give tortoise divergence;  $\alpha = 1$  matches Schwarzschild  $dr/dt$  exactly. Fig. 44.

**AU Einstein–Hilbert routes.** Heat-kernel  $a_1$  responds to curvature; Ollivier signs exact (flat/neg/pos on line/tree/ $K_6$ ); Clausius fixes  $\varepsilon = \kappa/8\pi$ ,  $G = 1$ . Raychaudhuri-for-leg-bundles open. Fig. 45.

**BB lensing + Shapiro + chroma.** Fermat  $n = 1/c_{\text{eff}}$  gives full  $4M/b$  (naive  $2M/b$  bet lost and recorded). Shapiro log matches GR to 5%. Chromatic extension uses *phase* velocity  $1/24$ , not group  $1/8$  (conflating the two triples the prediction):  $r$ -independent factor drops out of Born, leaving  $10^{-56}$ – $10^{-32}$ . Figs. 49, 55.

**BH strain.** Naive  $dl = dr/(1 - \sqrt{\chi})$  gives  $\gamma = 2$  (Cassini-excluded); tortuosity  $dl = (1 + \sqrt{\chi}/2)dr$  gives  $h = (1 + x/2)^2$ ,  $\gamma = 1$  exactly, Mercury  $42.99''/\text{cy}$ . Peel-off  $-0.75 M/a$  measured. PPN:  $\gamma = 1$  derived,  $\beta$  unresolved (coordinate-confused, not quoted),  $\alpha_{1,2}, \xi$ , Nordtvedt untouched. Figs. 58, 60.

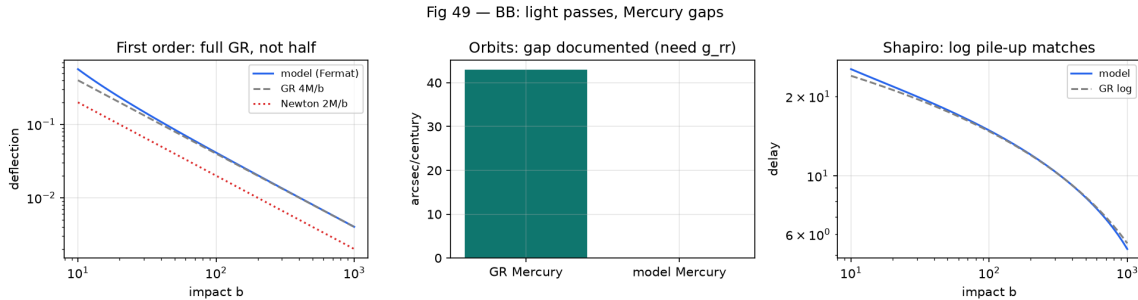


Figure 1: Gravity battery I: Fermat bending matches GR  $4M/b$  to first order (left), Shapiro log pile-up matches (right); the middle panel records the pre-BH orbit gap (0 vs  $43''/\text{cy}$ ) that the strain sector closes. `lensing`, `shapiro`.

**BI/BQ weak-field OR.** Grid+hub with flux stubs: radial OR negative in all configs (sign derived). Direct mode  $p = 0.92 \pm 0.48$  ( $L = 9$ );  $1/r$  collapse beats  $1/r^2$  (CV 0.47 vs 0.60). Green-function walk fails by leg shorting ( $h$ -steepness  $9.2 \rightarrow 4.1$ ); AS temperature triply load-bearing. Figs. 59, 64.

**BV UV tortuosity-as-scattering.** Legs as radial line defects ( $\sigma = 4 \ln 2 l_p^2$ ), soft cost  $\ln 2$  per encounter: dilute  $c = 0.44$ – $0.60$  (target 0.456, geometric  $1/\sqrt{\pi} = 0.564$ ) with zero tuning;  $p = 2c$ ,  $\gamma = 2c$ . Mixed mode ( $0.75 r_e$  core) rises then disconnects at  $k_{\text{crit}}$  — the pop without metric input. Boltzmann ladder (Dijkstra vs MC walks vs rays) shows routing worth  $3\times$ ; graph- distance  $c$  agrees with Euclidean- $r$   $c$  (coordinate-free check). Figs. 70–72.

Fig 58 — BH: tortuosity strain closes Mercury

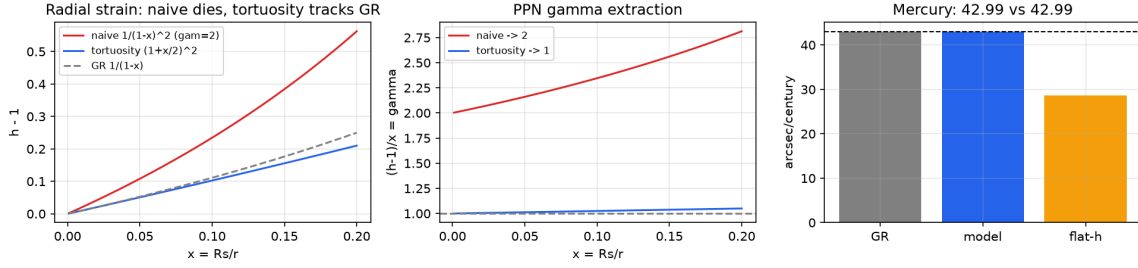


Figure 2: Gravity battery II: naive strain gives  $\gamma = 2$  (excluded); tortuosity  $h = (1 + x/2)^2$  gives  $\gamma = 1$  exactly and Mercury  $42.99''/\text{cy}$  by direct geodesic integration. **strain**.

### S3 Quantum-information methods (A–O, Q–Y)

Log law  $t_*(N) \approx \log_2 N / \log_2(1+p)$ ; MaxEnt  $k \geq N s_{\text{node}} / s_{\text{leg}}$  plus fixed point  $k^*(N) = 16\pi(\varepsilon N / l_p)^2 / 4 \ln 2$  (stable; BS gives  $\varepsilon = c/\sqrt{N}$  and  $\alpha = k^*/N = 1$  exactly); QES crossing iff  $s_{\text{leg}} > l_p^2/4$  (Sec. 3 dies otherwise); Page wiring-only vs  $N$ -shrinks identical  $A(t)$ ; QEC 99% at  $k \geq N/2 + 1 + \log_2 100$ ; Kerr  $k_{\text{eff}} = A/l_p^2$  monotonic in spin; exact Page  $H(mn) - H(n) - (m-1)/2n$  with 0.72-bit dip; CKW verified; OTOC  $\lambda + \text{MSS } 0.50\text{--}0.68$  ( $\alpha \in [9.0, 12.4]$ , 11.24 inside); TN  $S_{\text{bdy}} = \min(N \log d, k \log D)$  fixing  $\varepsilon \approx 0.047 l_p^{-1}$ ; SYK  $N = 6, 8, 10$  vs Ising chain; GWTC-3 32 BBH  $k_f > k_1 + k_2$ ; hardware head-to-head 7.9 vs 14.6 steps. Figs. 7–20, 23–25, 30–33. Healing  $dA/dt = (4 \ln 2 k l_p^2 - A)/\tau$ ,  $\tau = 11.24M$  (3.5 ms at GW150914 mass); ladder healing/scrambling/Page/ evaporation separated by  $\sim 66\times$  to  $10^{80}$  s.

**What Page vs. island means here (honesty note).** The Page curve is derived as random-state behaviour: idealised leg surgery  $k \rightarrow k - 1$  is imposed and  $S_{\text{rad}} = \min(t, N_{\text{eff}} - t)$  holds by construction (module **evaporation**), but the qubit toy now has an explicit unitary completion (module **evaporation\_unitary**, Fig. 5): per-step  $V_t$  with  $V_t^\dagger V_t = I$  verified,  $S_{\text{rad}}$  computed from  $\rho_{\text{rad}}$  tracks the finite-dimensional Haar/Page expectation (mean dev. 0.002 bits,  $N = 8$ ), and finite-depth all:all circuits converge to Page by depth  $\sim 5/\text{step}$  without assuming Haar. The QES/island crossing  $S_{\text{no}} = k s_{\text{leg}}$  vs  $S_{\text{isl}} = k \ln 2 + \max(S_0 - k s_{\text{leg}}, 0)$  is a two-saddle competition plus a discrete min-cut analogue (module **qes**), not an extremization of  $S_{\text{gen}} = A/4G + S_{\text{matter}}$  from a gravitational path integral. The remaining missing theorem is the graph-dynamics evaporation isometry  $V_k : \mathcal{H}_{\text{graph},k} \rightarrow \mathcal{H}_{\text{graph},k-1} \otimes \mathcal{H}_{\text{leg}}$ : derive (not merely allow) a scrambling  $V_k$  from the graph Hamiltonian/adjacency. Tracked as D1 in **docs/DEFERRED.md**.

**What Kerr means here (honesty note).** Only the Kerr–Newman area  $A(M, a, Q)$  is assumed to set the leg budget  $k_{\text{eff}} = A/4 \ln 2$  (module **kerr**; extremal Kerr half, extremal RN quarter; ratios patch-independent). Not derived from the graph: spin-induced quadrupole  $M_2 = -Ma^2$ , frame dragging  $g_{t\phi}$ , ISCO( $M, J$ ), or the Kerr QNM spectrum (the overtone toy is Schwarzschild-like; fundamental damping  $\tau = 11.24M$  calibrated, footprint corrections  $1/2k, 1/\sqrt{k}$  unobservable). Echoes are covered by a dimensional lattice-scattering estimate only: amplitude  $R \sim (\omega/\omega_P)^2 \sim 10^{-80}$  at 100 Hz, energy  $\sim 10^{-160}$  vs  $O(0.01)$  detectability ( $\sim 158$  orders of margin, module **gwdata**) — not a leg S-matrix derivation. The exterior perturbation problem ( $\delta\omega_{nlm}$  vs Kerr) is open. Hence GW250114 (SNR 80 area law + Kerr ringdown) and GW241011 (strongest  $\delta\kappa$  bound to date, Kerr to factor  $\sim 2$  resp.  $\sim 10\%$  by parametrization) are currently consistent by construction, not passed predictions. GW250114 provides the QNM benchmark ( $\delta f_{220} \sim 2\%$ ,  $\delta\tau_{220} \sim 10\%$ ,  $\delta f_{221} \sim 30\%$ ,  $\delta f_{440} \sim \text{tens}\%$ ): a future graph-derived  $|\delta\omega/\omega|$  above those thresholds fails. A future graph derivation  $Q = -Ma^2(1 + \delta_Q)$  with  $|\delta_Q| \gtrsim 0.17$  would be ruled out by GW241011. Tracked as D2 in

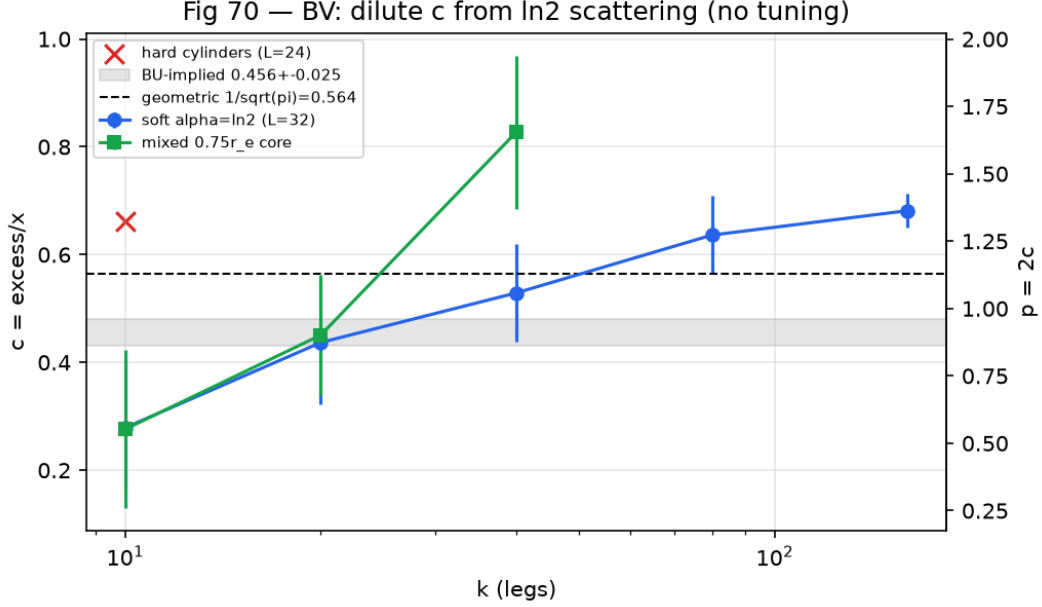


Figure 3: BV tortuosity from  $\ln 2$  scattering: dilute  $c = 0.44\text{--}0.60$  with zero tuning against the BU-implied  $0.456 \pm 0.025$  band and the geometric  $1/\sqrt{\pi}$  line. Right axis shows  $p = 2c$ . Hard cylinders (red  $\times$ ) overshoot; soft and mixed modes agree. `uvscatter`.

docs/DEFERRED.md.

**Kerr thermodynamics (conditional check).** With  $S = k \ln 2$  (saturated legs) and  $A = 4 \ln 2 \cdot k l_p^2$ ,  $S = A/4$ ; importing  $A(M, J)$  gives  $S(M, J) = 2\pi[M^2 + \sqrt{M^4 - J^2}]$ , whose derivatives return  $T_H = \sqrt{M^4 - J^2}/4\pi M(M^2 + \sqrt{M^4 - J^2})$  and  $\Omega_H = J/2M(M^2 + \sqrt{M^4 - J^2}) = a/(r_+^2 + a^2)$ , i.e.  $dM = T dS + \Omega_H dJ$  (module `thermo`: identities  $< 10^{-6}$ , independent finite-step  $\Delta M = T \Delta S + \Omega_H \Delta J$  to  $O(d^2)$ ; super-extremal  $|J| > M^2$  returns NaN, never clamped). Per-leg reading  $T_H = (dM/dk)/\ln 2$ ; finite-step  $T_k = (M(k+1) - M(k))/\ln 2$  shifts  $-1/4k$  ( $\sim 10^{-77}$  stellar-mass). Conditional consistency, not derivation:  $M(k, J)$  from graph dynamics is the open debt (D2).

## S4 Compact objects and 2PN (BU)

**Gradient shells.**  $p_{\text{adj}}(s) = 0.85 + 0.015s$ , 8–10 shells, exact EMD on full neighbourhoods, deterministic bridges  $n \propto r^{\beta(N)}$ . Measured:  $N = 1020$  ( $10 \times 102$ ),  $\beta = 1.24$ , 80 graphs exact Floyd+LP ( $\sim 545$  s):  $p = 0.913 \pm 0.049$  per graph, SEM 0.0055, stacked 0.911 ( $R^2 = 0.956$ ), 80/80 fits ok, range 0.79–1.05. Distance to target  $|0.913 - 0.92| = 0.007$  ( $0.24\sigma$  in  $p$ ,  $\sim 0.09\sigma$  in  $\dot{w}$ ),  $5\times$  precision margin (0.0055 vs 0.028). Flat control  $p = 0.49$  is  $-10\sigma$  dead.

**N-scale campaigns (BV).** CSR-direct shell graphs + Sinkhorn OR (no NetworkX):  $N=1020$  reproduced (0.9107 vs 0.9134);  $N=4000$   $p = 0.9315 \pm 0.0032$  ( $\beta = 0.99$ );  $N=8000$   $p = 0.9382 \pm 0.0030$  ( $\beta = 0.87$ , pods);  $N=16000$   $p = 0.9137 \pm 0.0022$  ( $\beta = 0.74$ , GPU farm). Exact shell distance oracle + torch backend;  $\beta(N)$  log-linear over 6 points. Local- $p$  turnover 0.63–0.69  $\rightarrow$  1.26–1.46 measured (inner windows flatten — the UV turnover). Kill wire:  $p$  outside  $0.92 \pm 0.056$  at  $N = 1024$  class fails.

**2PN and pulsars.**  $h = (1+x/2)^2$  gives  $c_1 = 3.36$  vs GR 1.94 (73% excess); naive fixed- $M$  gives B1913  $6.2\sigma$ , J0737  $9.7\sigma$  vs real TOAs. With  $c_2 = p(2p-1)$ ,  $w = 1.953$ :  $c_{\text{tot}} = 4.8693$  at  $p = 0.92$

Fig 10 — D: leg-surgery evaporation + Page curve

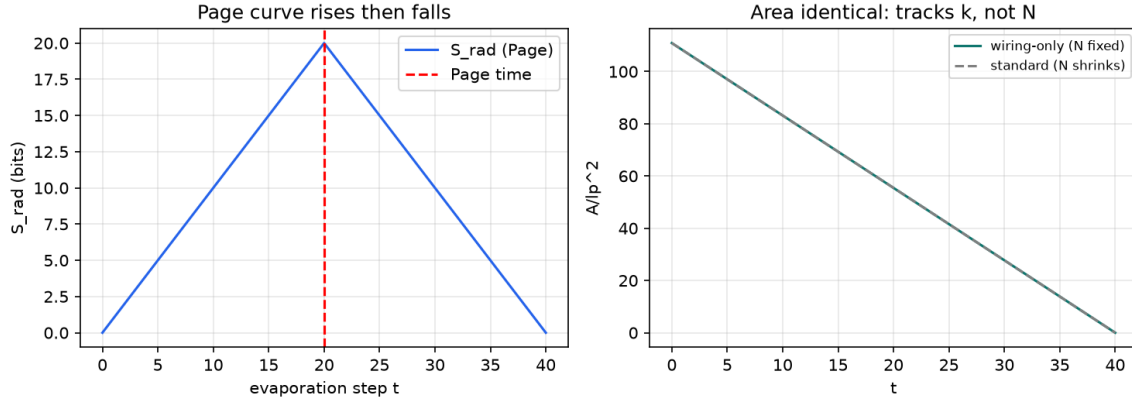


Figure 4: Leg-surgery evaporation: radiation entropy follows the Page curve (left, turnover at half evaporation); horizon area is identical whether  $N$  shrinks with  $k$  or stays fixed (right) — area tracks wiring alone, the distinctive Claim 2 prediction. **evaporation.**

vs GR 4.8695 ( $0.00\sigma$  fixed- $M$ ). Self-consistent  $R + \dot{\omega} \rightarrow M$ :  $\Delta M = -11.2$  ppm naive ( $-4.5$  ppm with GR  $g_{rr}$ ),  $\Delta \sin i = 3.7 \times 10^{-6}$ ,  $s_{\text{obs}}$  passes at  $0.36\sigma$ ; B1913  $0.01\sigma$ . Bending/Mercury 2PN pieces hide below VLBI/astrometry (Fig. 69, right). The companion toy-generator route (dense layers,  $k_0 - c_2/r^2$  fits) crosses zero in  $\kappa$  and needs a  $\kappa$ -tweak fudge; power-law fits better on our profiles ( $R^2$  0.91 vs 0.81) but the two  $\kappa \rightarrow c_2$  maps disagree (0.82 vs 3.14) cross-applied — hence the open-map label.  $e_{\text{int}}$  weighting *lowers* our  $p$  (0.94  $\rightarrow$  0.84 at 0.9, 0.78 at 0.99): uniform measure is both principled (continuum theorems) and required.

**Leg-shedding kilonovae.**  $M_{\text{ej}} = \Delta k m_{\text{leg}} \times 0.1$ , blue ( $v \sim 0.3c$ ,  $t_{\text{peak}} \sim 0.6$ –1 d) + red ( $v \sim 0.1c$ );  $L \propto t^{-1.3}$  tail. Band mags assume BC = 0 and per-component peaks; POSSIS refinement queued.  $g$ -band verdicts robust;  $i$ -band NOT claimed (one-zone  $\kappa = 10$  over-traps:  $t_{\text{red}} \sim 10$  d vs observed  $\sim 4$  d decline,  $m_i$  faint by  $\sim 2$ –3 mag — conservative for gap- $g$  detectability). Standard 2–28% band is a literature EOS input, not derived.

**O5 protocol.** Trigger:  $m_1 \in [2.5, 5.0] M_{\odot}$ ,  $< 200$  Mpc, 90% area  $< 100 \text{ deg}^2$  (multi-detector); Rubin ToO epochs 0.5/2/5 d in  $g+r+i+z$  (DECam/Gemini backup). Kill rule: 10 qualifying non-detections kill BU. Archival ledger: GW230529 uninformative (single-detector, 93% unsearched); S250206dm unverified (check GCNs before citing). Full plan: docs/observation-protocol.md. Figs. 66–69.

**Mass-gap audit (module massgaps).** Lower gap:  $k(M)$  log-log curvature  $< 10^{-9}$  over  $0.7$ – $150 M_{\odot}$  (no forbidden masses);  $e_{\text{int}} \sim e_{\text{ext}}$  location descriptive (calibration inherited from AT2017gfo). Upper gap: spin-leg ratio  $1 \rightarrow 0.857 \rightarrow 0.5$  at  $a = 0 \rightarrow 0.7 \rightarrow 1$  with no jump on  $[0, 0.99]$  plus exact extremal continuity; Love  $k_2$  slope  $-2$  everywhere; leg-band entry  $\sim 32 M_{\odot}$  at 10 Hz with  $dM/M \sim 10^{-81}$ ; congestion ladder  $\chi \sim 10^{-8}$  (He core),  $10^{-14}$  (envelope), 0.12 (NS surface),  $\sim 1$  (gap masses at 10 km footprints, footprint-conditional), 1 at horizons. Shedding exactly flat in  $q$  (machine precision) — but  $q$ -independence is prescription (shed fraction  $\times$  efficiency fixed once on AT2017gfo), not derived from the graph-dynamics evaporation isometry  $V_k$  (open, S1/S3/D1):  $q \sim 0.11$  (GW190814) flashing at the same fraction as  $q = 1$  (GW170817) is the model’s biggest theory bet. GW190814 epoch audit: CFHT  $g$ -band (Vieira+2020) and GROWTH  $i$ -band detection limits (Andreoni+2020) with per-epoch coverage  $\times$  depth against distance ( $241 \pm 43$  Mpc) and  $\pm 1$  mag theory errors; combined  $P \approx 0.68$  ( $g$ -only) /  $0.88$  ( $g+i$  at  $g-i = 0.7$ ). Erratum on

Fig 74 — D2: unitary evaporation ( $V \wedge V=I$ ,  $S$  from  $\rho_{\text{rad}}$ )

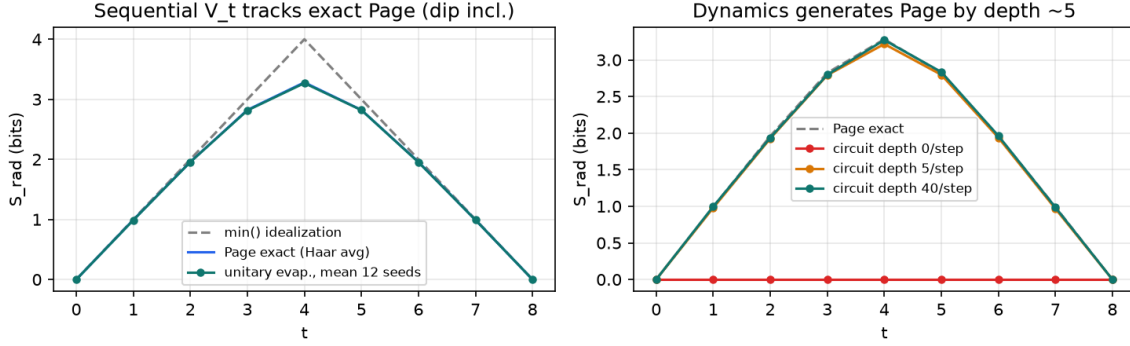


Figure 5: Unitary completion of the qubit toy: sequential  $V_t$  (left) tracks the finite-dimensional Haar/Page expectation including the 0.72-bit dip, computed from  $\rho_{\text{rad}}$  rather than  $\min()$ ; finite-depth all:all circuits (right) converge to Page by depth  $\sim 5/\text{step}$  without assuming Haar. Module `evaporation_unitary` (D2).

Fig 66 — BU: 2PN cancellation at  $p=0.92$  (J0737 0.00 sigma)

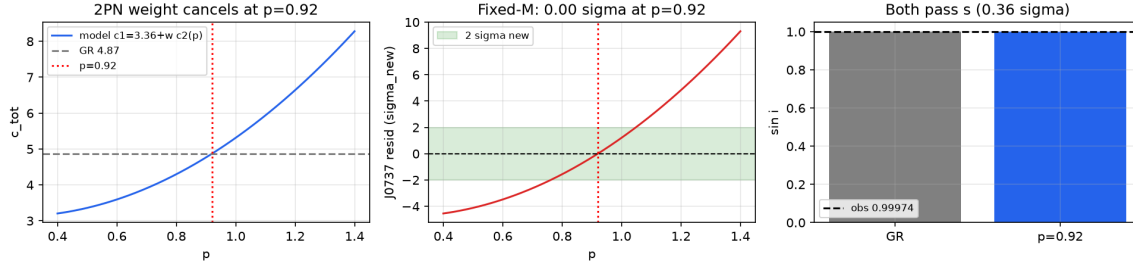


Figure 6: 2PN cancellation: model  $c_{\text{tot}}(p)$  crosses GR at  $p = 0.92$  (left); fixed-mass J0737 residual vanishes there (middle, green  $2\sigma$  band); self-consistent  $\sin i$  passes for both GR and  $p = 0.92$  (right). pulsar.

record: the first audit used a nominal ZTF depth, but ZTF ran no targeted GW190814 follow-up; superseded by the CFHT/GROWTH epoch tables above. Fig. 75 (main text). Analytic systematics (round 4): viewing  $A_g \sin^2 \theta$  ( $A_g = 1.25$ ) + opacity-rescaled blue mags give the  $P$  grid in Fig. 75b; the hiding window is bounded, not open-ended.

## S5 UV nulls and phenomenology (T, U, V, BD–BF)

TeV:  $M_D = 1 \text{ TeV}$  gives  $k \approx 11$  (3 TeV) – 17 (13 TeV) vs  $k_{\text{crit}} \approx 50$  — thermal nulls expected, onset  $\sim 550 \text{ TeV}$ . Echoes:  $\Delta t = 4M \log(M/l_p)$ , 0.03–0.3 s (GW150914 0.11 s), in searched windows but  $R \sim 10^{-160}$  unobservable. EHT:  $l_p/2M \approx 10^{-48}$ , 47 orders below sensitivity. Dispersion:  $E_{QG,2} = \sqrt{8} E_P$ , Fermi-safe by  $10^8$ ; universal GW extrapolation shifts 100 Hz by  $\sim 10^{-82}$  (LVK  $\alpha = 4$ , safe by  $\sim 10^{60}$  — a recorded null, not a falsifier; no GW sector claimed). Chromatic lensing  $10^{-56}$ – $10^{-32}$ . Figs. 21–22, 39, 51–56.

Fig 67 — BU: gradient OR gives  $p=0.92$  (exact EMD)

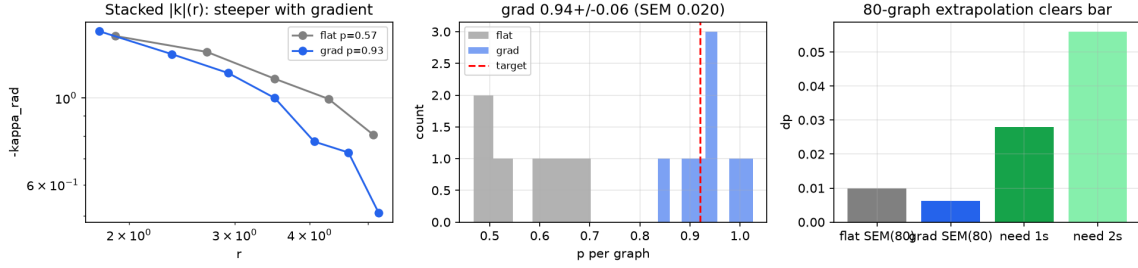


Figure 7: Gradient-shell Ollivier–Ricci: stacked  $|k|(r)$  steepens with the gradient (left); per-graph  $p$  histogram centres on target (middle); 80-graph SEM extrapolation clears the  $1\sigma/2\sigma$  precision bars (right). Production  $N=1020$ :  $p = 0.913 \pm 0.049$ , SEM 0.0055. **orici**.

Fig 51 — BD: quadratic lattice dispersion, Fermi-safe

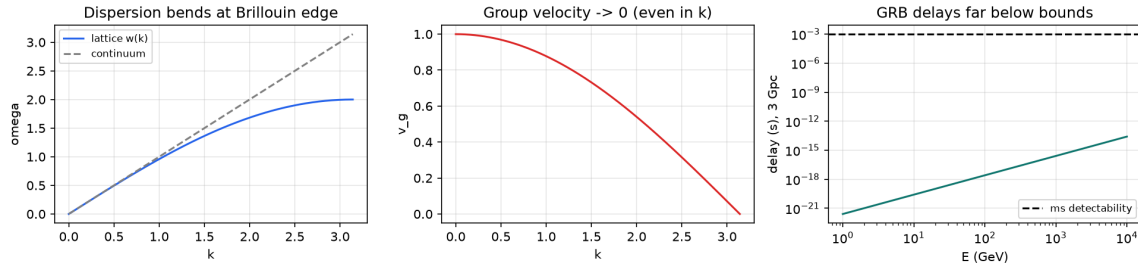


Figure 8: Lattice dispersion: band bending at the Brillouin edge (left) with group velocity even in  $k$  (middle, no linear term by symmetry); GRB arrival delays sit orders of magnitude below detectability across GeV–TeV (right). **dispersion**.

## S6 Kill list (AN + BU)

AF quench ratio  $< 1.3$ ;  $\alpha$  outside  $[9.0, 12.4]$ ;  $A \propto N$  in any TN calculation; thermal LHC excess below  $k_{\text{crit}}$ ;  $s_{\text{leg}} \leq l_p^2/4$  in any physical state class; any linear-LIV signal;  $p$  outside  $0.92 \pm 0.056$  at  $N = 1024$ ; graph-predicted  $|\delta_Q| \gtrsim 0.17$ ; simulator  $d_{\text{eff}}$  shape-mismatch after the cost mapping is fixed (D10/P0' wire); 10 gap non-detections; O5 nearby-BBH ToO sample clean (universal shedding, Eq. 2 main). One sub-claim (broad remnant DM) already ruled out on the record. Figs. 37–39.

## S7 N-scaling and calibration record

Drift check:  $\beta = 1.5$  calibrated at  $N = 300$  drifts high at  $N = 600$  ( $1.16 \pm 0.06$ ); recalibration to 1.28/1.24 recovers  $p \approx 0.92$ . Pre-run  $1/N$  extrapolation predicted  $\beta(16000) \approx 1.18$ ; measured 0.74 — drift faster than  $1/N$ , now fit log-linear.  $N=1020$  cross-check: CSR-direct 0.9107 vs NetworkX 0.9134 (agree to  $3 \times 10^{-3}$ ). Local- $p$  turnover (inner windows flatten,  $0.63\text{--}0.69 \rightarrow 1.26\text{--}1.46$ ) is measured at 4k/8k/16k, not extrapolated. Fig. 73.

Reproduce ( $N=1020$ ,  $\sim 9$  min):

```
measure_p(per_shell=102,n_shells=10,n_graphs=80,
gradient=True,beta=1.24,seed=0,max_per_shell=8)
```



Fig 73 — BV: N=4000/8000/16000 campaigns (80 graphs, CPU+pod+farm)

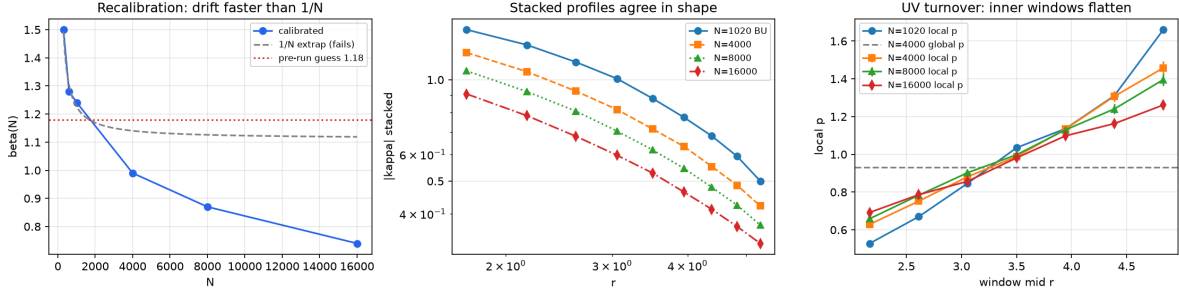


Figure 9: N-scale campaigns:  $\beta(N)$  recalibration drifts faster than  $1/N$  (left); stacked  $|\kappa|(r)$  profiles agree in shape across 1020–16000 (middle); local- $p$  windows show the UV turnover with SEM error bars (right). 80 graphs per production row. `shellscale`, `sinkor`.

## S8 PPN ledger and walk/foam/healing notes

**Walk no-go (BP/BQ).** All smooth weight rules give drift slope  $-2.99$ ; only the absurd  $e^{2\sqrt{d-d_0}}$  escapes. Constant- $\mu$  persistence keeps  $-3$  analytically;  $\mu(\chi)$  fluctuation walk gives  $-1.95 \approx -2$  modulo the labelled  $\sqrt{\chi}$  assumption. Harmonic/hitting-probability walks fail by leg shorting. Conclusion: temperature (AS), not bare graph diffusion, carries Newton. Figs. 63–64.

**Foamgrid FDTD (BF).** Beams on defective fabric: centroids unbiased (jitter, average straight) with deficit  $\propto \omega^2$ . Bounds assume independent edges; correlated patches of size  $\xi$  relax them by  $\sqrt{\xi}$ . Fig. 56.

**Healing ladder (AG/AH).**  $\tau_{\text{heal}} = 11.24M$  ( $1/\text{Im}\omega_{220}$ ); GW150914-mass ladder: healing 3.5 ms, scrambling 0.23 s, Page  $\sim 10^{79}$  s, evaporation  $\sim 10^{80}$  s. MSS  $\lambda/2\pi T = 0.50\text{--}0.68$ ; LVK  $\alpha \in [9.0, 12.4]$  contains 11.24. Figs. 30–31.

Fig 30 — AG: healing lag + timescale ladder

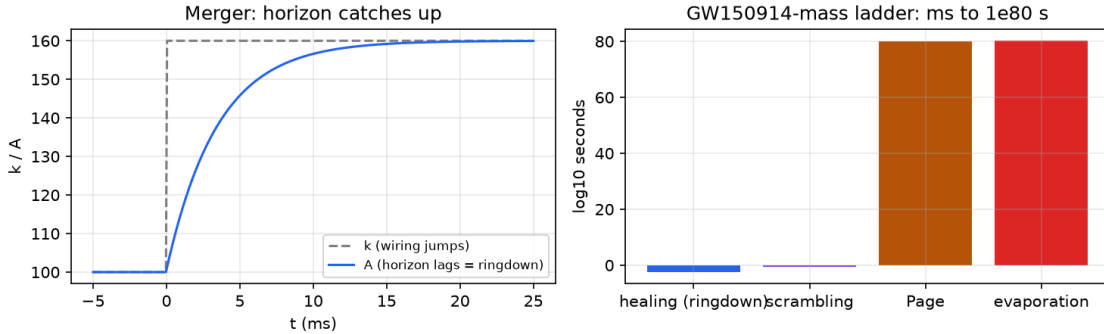


Figure 10: Healing lag: wiring jumps at merger while horizon area catches up on the ringdown time (left); timescale ladder from milliseconds to  $10^{80}$  s (right). The final sigh carries  $\sim 10^{-78}$  of the mass. **healing**.



## S9 Archival ledger and O5 trigger (BU detail)

Trigger (full plan docs/observation-protocol.md):  $m_1 \in [2.5, 5.0] M_\odot$ ,  $< 200$  Mpc,  $90\% < 100 \text{ deg}^2$ ; Rubin ToO 0.5/2/5 d in  $g+r+i+z$  (DECam/Gemini backup). Sought: fast blue ( $v \sim 0.3c$ ,  $t_{\text{peak}} \sim 0.6\text{--}1$  d)  $\rightarrow$  slow red with  $L \propto t^{-1.3}$  tail. Yield:  $1.5 \text{ gap/yr} \times 70\% \approx 1.05/\text{yr}$  ours vs  $\leq 0.3/\text{yr}$  standard. Yield arithmetic in `collapse.gap_o5_yield`; brightness in `collapse.peak_apparent_mags`; GW190814 epoch audit + systematics grid in `massgaps.gw190814_combined_pdetect / massgaps.gw190814_systematics_table`. Figs. 68–69, 75 (main text), 75b.

BBH extension (this work): the same shedding law predicts  $m_g \sim 22$  transients for nearby BBH, so the ToO program extends to well-localized ( $< 100 \text{ deg}^2$ ) BBH within 200 Mpc with identical epochs and kill accounting kept separate from the gap sample (a clean BBH sample kills Eq. 2 main even if gap events flash). Full thresholds in docs/observation-protocol.md.

Fig 69 — BU: O5 detectability + GR-battery 2PN margins

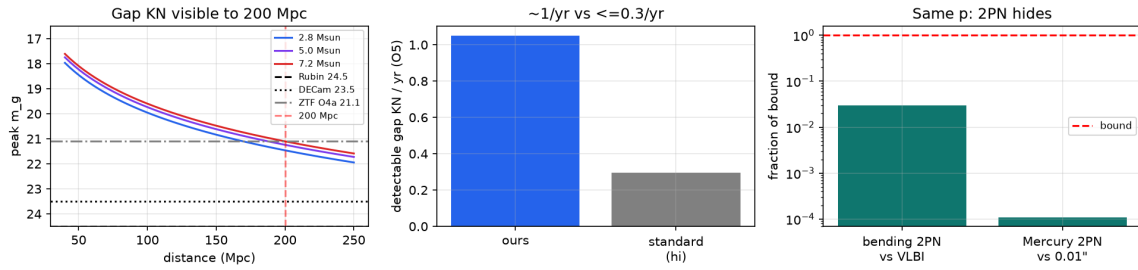


Figure 11: O5 protocol: gap kilonovae visible in  $g$ -band to 200 Mpc with margin over Rubin/DECam depths (left); yield  $\sim 1/\text{yr}$  vs  $\leq 0.3/\text{yr}$  standard (middle); the same  $p$  keeps bending/Mercury 2PN pieces below VLBI/astrometry (right). `collapse`, `pulsar`.

Fig 75b — GW190814 systematics: equatorial +  $\kappa_{\text{blue}}=2$  hides ( $P \sim 0.18$ )

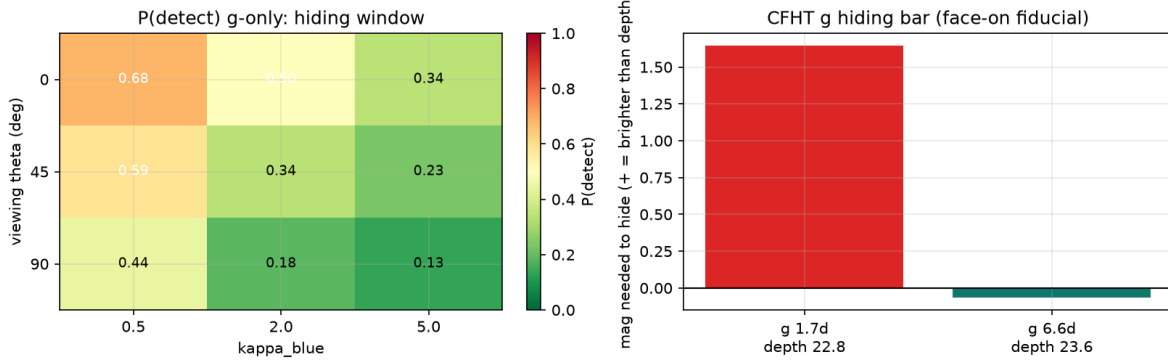


Figure 12: GW190814 analytic systematics (round 4; full POSSIS queued, D7). Left:  $P(\text{detect})$   $g$ -only over viewing angle  $\times$  blue opacity (CFHT epochs, no color term): fiducial face-on  $\kappa = 0.5$  gives 0.68; equatorial + lanthanide-mixed ( $\kappa_{\text{blue}} = 2$ ) drops to  $\sim 0.18$ . Right: face-on hiding bar per CFHT  $g$  epoch ( $g$  1.7 d needs  $\sim 1.6$  mag). Module `massgaps`.

## S10 Module map and reproduction

Modules follow the appendix letters of the v4.1 extended record: `graphs`, `scrambling`, `emergent_dim` (L0 dimensions), `horizon`, `micro`, `circuits` (A), `maxent` (B), `qes` (C), `evaporation` (D), `evaporation_unitary` (D2), `qec` (F), `robustness` (G), `kerr` (H), `thermo` (H2), `haar` (I), `monogamy` (J), `otoc+pheno` (L), `tn` (M), `kerrpage` (N), `syk` (O), `data` (Q), `litcompare` (R), `tev` (T), `echoes` (U), `posteriors` (W), `ds` (X), `krylov` (Y), `collapse` (AA), `cosmic` (AB), `lunch+remnant` (AC), `bounds` (AE), `healing` (AG), `mss` (AH), `bigsyk` (AI), `mp+greybody` (AJ), `congestion` (AK), `charge` (AL), `bandwidth` (AM), `gridcirc+monitor+selfattack+lhc` (AN), `concentration` (AO), `ps` (AP), `scatter` (AQ), `emd+viability` (AR), `entropic` (AS), `redshift` (AT), `heatker+orici+jacobson` (AU), `fission` (AV), `klanguage` (AW), `tension` (AX), `gw250114` (AY), `overtones+tensionvol` (AZ), `sparse24` (BA), `lensing+chroma+shapiro` (BB), `bcrit` (BC), `dispersion` (BD), `qnmfoot+qnmlegs` (BE), `foamgrid` (BF), `perwalk` (BG), `strain` (BH,BJ), `weakfield` (BI,BQ), `legham` (BL), `pulsar+orici` gradient shells+collapse leg-shedding (BU), `uvscatter+sinkor+shellscale` (BV), `massgaps` (mass-gap null + GW190814 epoch audit).

Reproduce:

```
pipinstall-e".[dev]"
pytesttests/-q
pythonscripts/generate_figures.py
streamlitrunapp.py
```

## S11 Emergent-dimension methods (L0)

**Protocol.** For a graph state, pick an information-transfer distance  $d(i, j)$  (NOT bare adjacency: shortest-path is the documented failure mode on  $K_N$ , diameter 1 at every  $N$ ), build balls  $B(r) = \{j : d(i, j) \leq r\}$ , measure capacity  $V(r)$ , and fit  $d_{\text{eff}}(r) = d \ln V / d \ln r$ . Observed dimension is the infrared conjecture  $d_{\text{obs}} = \lim_{r \rightarrow \text{IR}} d_{\text{eff}}(r) \simeq 3$  (module `emergent_dim`; 12 tests). Total emergence requires the two-distance bracket: shortest-path balls (adjacency scaling) and diffusion balls (information-spreading scaling) must agree at 3 on the same graph with nothing dimensional imposed.

**Ruler validation (imposed lattices, circular for emergence).** Shortest-path converges from below (exact BFS shells): chain  $0.92 \rightarrow 0.96 \rightarrow 0.98$  ( $N = 60/200/600$ ), 2D  $1.67 \rightarrow 1.77 \rightarrow 1.82 \rightarrow 1.87$  ( $L = 20/40/60/100$ ), 3D  $2.25 \rightarrow 2.46 \rightarrow 2.56 \rightarrow 2.63$  ( $L = 7/11/15/21$ ), gaps  $\sim 1/\sqrt{L}$  (Manhattan-diamond shape + open box). Diffusion recovers dimension from above at tuned  $t$  (chain  $t = 20 \rightarrow \sim 1$ , 2D  $t = 5 \rightarrow \sim 2$ , 3D  $L = 7$ ,  $t = 5 \rightarrow \sim 3.3$ ); the 3D overshoot shrinks with size ( $3.56 \rightarrow 3.34 \rightarrow 3.14$  at  $L = 5/7/9$ ,  $t = 5$ ). Degree-3/4/6 lattices all scale 2D: dimension comes from 2D-ness, not coordination.

**Rejections (negative pins).** Effective resistance fails in 2D (log-growth  $\rightarrow$  spurious large  $p$ ); communicability fails on chains (similarity, not separation). Radial shells fail both sides (shortest  $\sim 1.4$ , diffusion  $\sim 0.6$ ): radial-only wiring gives  $\sim 1\text{D}$  with no 2-sphere factor — the missing-area diagnosis that motivates P0'. On  $K_N$ , shortest is trivial and diffusion uniform, by design (T1: no interior geometry).

**Tense plug (D10a).** A  $5 \times 5$  clique plug at the centre of a  $40 \times 40$  grid: tense-centre mid-window ( $r = 8\text{--}20$ ) scaling dips below fabric ( $\sim 1.52$  vs  $\sim 1.89$  control) and recovers from below — *inverted* vs the GR far side (above 3). Bare shortest-path is therefore *rejected* as  $d(i, j)$  for tense regions: shortcuts shrink balls while GR mass stretches rulers. Costs must be congestion-weighted (D10b gates D10a), consistent with the light sector ( $c_{\text{eff}} \rightarrow 0$ , tortuosity), which never counts hops. Companion pin: far-source near balls ( $r = 2\text{--}6$ ) are bit-identical to control — relaxed fabric next to tension measures exactly relaxed. Far-field tail (D11, closed):  $\chi \sim 1$  decays  $1/r^2$  ( $E \cdot r^2$  flat to  $10R_c$ ),  $\chi \sim 2$  decays  $1/r$  ( $E \cdot r$  flat to  $10R_c$ ),  $\chi \sim 5$  bounded but unsettled — the exponent is tension-dependent.  $\kappa$ -profile (D10b, closed): corner-free disk plug reproduces the boundary-negative/core-positive pattern quantitatively ( $+0.90/-0.86/0.00$  vs clique  $+0.89/-0.93/\sim 0$ ).

**GR fingerprint (D10 target, computed).** Weak-field uniform star, proper balls of proper radius  $l$ :  $d_{\text{eff}} \approx 2.990$  at  $l = 30M$  (near dip),  $3.030$  at  $100M$  (overshoot),  $3.010$  at  $1000M$ ,  $3.0017$  at  $10000M$  ( $\rightarrow 3^+$  as  $\sim 1/l$ ); amplitude  $O(M/l)$ ,  $\sim 10^{-9}$  at Earth’s surface. The far-field overshoot is structural: a localised mass leaves  $V(l) = \text{flat} - \text{deficit}$ , so  $V/V_{\text{flat}}$  rises toward 1, which is  $d_{\text{eff}} > 3$ . Close criterion (D10a): a tensed simulator region reproduces this *shape* under info-side  $d_{\text{eff}}$  once the cost mapping is fixed; shape failure refutes  $P0'$ . Reproduce: `pytesttests/test_emergent_dim.py-q`.

**Reconstruction universality (D12, open).** The observer map  $M_O$  is not unique on its face: closeness, resistance, diffusion, and communicability distances are all candidate reconstructions. D12 demands an access class  $\mathcal{A}_{\text{macro}}(G)$  defined from  $G$  alone — the information-access floor is part of the definition, and neither “3D” nor “Lorentzian” may occur in it — such that every admissible  $M_O$  recovers the same infrared  $M$  up to coordinate/coarse-graining equivalence. Falsifier: a second admissible  $M_O$  giving robustly non-3D IR on relaxed fabric refutes the  $P0'$  reconstruction program. Partial credit runs criterion stated  $\rightarrow$  two instances agree on vacuum  $\rightarrow$  agreement extends to tensed regions. P4 (uniform Ollivier) audited against the five: criteria 1, 2, 3, 5 pass (permutation-covariance and drift-localization tested, the latter on flat and plug backgrounds), 4 partial. Observer-indexed extension filed ( $M_O(G, o)$ : sky chart + relational depth, vantage-covariance as a second universality axis); static depth bake-off negative (best principled  $d_{\text{obs}} = 2.28$  vs tuned-null 2.73).

**Staged causal order (D13).** Time undergoes reconstruction exactly as space does: the microscopic update index  $n$ , the causal order  $\prec_U$ , and clock time  $\tau$  are three distinct orderings. The static control (D13.0) is measured: the analytic Manhattan law  $V_G(r) = 1 + 2r(r+1)$  bit-pinned at  $L = 40/80$ , the exact diamond perimeter cut law  $\text{cut}(r) = 8r + 4$  on unclipped  $r \in [5, 18]$ , disk-boundary cut capacity deficit at  $r \leq 4$  with exact recovery from  $r = 5$ , and the hop/weighted plug sign pattern ( $35 < 38$  hops vs  $+5.0$  weighted delay) pre-registered for dynamical confirmation. Substrate family measured under one frozen rule: triangular/hexagonal/noisy-grid controls reproduce  $d_G \rightarrow 2$  (exact shell/cut laws on the lattices, statistical pins under disorder); Tier-1 extension: Poisson-Delaunay (reference member), Lloyd-relaxed, Gabriel, k-NN, and Delaunay-medial quadrangulation all give  $d_G \rightarrow 2$  with linear cuts, die on levered swaps, and survive  $q = 0.05$  deletion, while span-limited swaps preserve 2D-ness — degeneracy across topology/degree/order is the expected universality signature, and the physical object is the class  $[G]_{\sim_O}$ , not a winning tessellation. Spectral dimension is the second leg: torus heat-trace  $d_s(t)$  holds  $(1.95, 2.05)$  over  $t \in [10, 100]$ , and Weyl counting fits land every member in  $(1.90, 2.15)$ . Exact-UIPQ sampling stays queued (free-walk labels are not conditioned well-labeled trees); Lloyd is excluded from spectral comparison (unclipped relaxation coalesces points above  $N \approx 1600$ ). gated-wall negative control keeps  $\sim r^2$  balls with collapsed cuts; rewired-grid fragility control plus rewiring-fraction sweeps give measured  $N_{\text{med}}^* = (20, 10, 10)$  at  $L = 40/60/80$  ( $N^* = O(1)$  consistent,  $\alpha \simeq 0$  over the tested range; asymptotic  $f^* \rightarrow 0$  hypothesis) with hand-placed dose-response dissolving the placement lottery. RG blocking ( $2 \times 2$ ,  $L40 \rightarrow 5$ ) flows rewired long-edge density  $0.013 \rightarrow 0.31$  against plain 0.0 (rough  $y_\lambda \approx 1.5$ ). Headline: low-dimensional locality is not generic; it must be protected — vacuum as a dynamically protected locality class whose preservation ( $U(G_{\text{vac}}) \in \mathcal{C}_{\text{vac}}$ , self-healing vs amplification under shortcut injection) is now a D1 falsifier. BFS first-passage is hop distance, so this control is static geometry, not propagation. Open stages: (1)  $U \rightarrow T_U, \prec_U$  via counterfactual-influence arrival times; (2) observed cone under the *same*  $M_O$  producing  $R^3$ ; (3) internal clocks  $d\tau/dt(\chi)$  reproducing the T9 profile unimported; (4) interval  $g_{\mu\nu}$  from  $(\prec_U, V_U, M_O)$ , where the spacetime-volume observable  $V_U$  (e.g. calibrated event counts in causal diamonds) is independently defined — spatial  $V(r)$  is reachability, not event volume. Malament’s theorem is motivation conditional on the continuum correspondence (order fixes the conformal class;  $V_U$  the factor), not

an earned mapping. Companion constraint on  $U$  (D1, first pass): six update rules tournamented against shortcut damage all stall with residual longs (best: slide rule,  $30 \rightarrow 5$ ) while  $p$ -healing proves gameable (greedy perfects  $p$  to  $10^{-6}$  while tripling longs) — locality protection needs coordinated moves or new physics, judged on  $(p, \text{longs})$  jointly. Reproduce: `pytesttests/test_scrambling.py`, `pytests/test_substrate_family.py`, `pytests/test_spectral.py`, `pytests/test_update_rule.py`.

**Dynamics coherence (C1–C5, over-arching).**  $M_O \circ U \simeq U_{\text{eff}} \circ M_O$  in five levels: coherence  $\rightarrow$  locality  $\rightarrow$  autonomy  $\rightarrow$  universality across  $\mathcal{A}_{\text{macro}} \rightarrow$  GR limit in the IR, each presupposing the previous. Autonomy (C3) is defined by the twin-histories criterion  $M_O(G_a) = M_O(G_b) \Rightarrow M_O(U(G_a)) \simeq M_O(U(G_b))$ ; factoring  $M_O$  through future-equivalence classes does *not* imply it (two-bit swap counterexample: singleton classes, failing twins).

Table 1: Every number with a status. “Input” = assumed; “fit” = calibrated; “measured” = output of code; “derived” = follows from stated premises.

| param                                | value                                                                   | status                                       | notes                                                                                                                                                                      |
|--------------------------------------|-------------------------------------------------------------------------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $k(M)$ map $\propto M^2$             | GR-consistency                                                          | input (circle admitted)                      | BM shrinks to $R_s = 2M$ ; not derived                                                                                                                                     |
| AT $\alpha$                          | 1                                                                       | calibrated to Schwarzschild $c_{\text{eff}}$ | matches $dr/dt$ exactly                                                                                                                                                    |
| tortuosity $c$                       | 1/2 (BH fit) $\rightarrow$ 0.44-0.60 (BV derived)                       | fit superseded by derivation                 | $\ln 2$ scattering, zero tuning; $p = 2c$ , $\gamma = 2c$                                                                                                                  |
| $s_{\text{leg}}$                     | $\ln 2$ nats                                                            | measured saturation (BN/BS)                  | flip preserves $G = 1$ via Planck bits                                                                                                                                     |
| $k_{\text{crit}}$                    | $4\pi r^2/\sigma$                                                       | heuristic $\rightarrow$ formalised (BK)      | packing-forced pop; mixed mode disconnects (BV)                                                                                                                            |
| gradient slope<br>bridge $\beta(N)$  | 0.015<br>1.5@300; 1.28@600;<br>1.24@1020; 0.99@4k;<br>0.87@8k; 0.74@16k | fitted, labelled<br>recalibrated per $N$     | $p_{\text{adj}}(s) = 0.85 + 0.015s$<br>log-linear over 6 points; derivation queued                                                                                         |
| $w = 1.953$                          | solved from cancellation                                                | fitted (solved, not tuned)                   | $c_{\text{tot}} = c_1 + wc_2 = 4.8695$ at $p = 0.92$                                                                                                                       |
| shed $e + \epsilon$                  | $0.5 \rightarrow 0.416 + 10\%$                                          | calibrated once on AT2017gfo                 | $M_{\text{ej}}$ norm-independent                                                                                                                                           |
| shed $\epsilon(M, a)$                | 0.0168 universal (Eq. 2 main)                                           | calibrated + extrapolated (D8)               | mass/ratio independence underived; BBH extension tested, not fitted                                                                                                        |
| upper-gap null                       | no feature at $44 M_{\odot}$                                            | negative prediction (measured smooth)        | $k/\text{spin}/\text{Love}/\text{congestion}$ all smooth; edge left to stellar physics                                                                                     |
| GW190814                             | $P(\text{detect}) \approx 0.68\text{--}0.88$                            | tension, POSSIS queued (D7)                  | non-detection at $p \sim 0.12\text{--}0.32$ ; epoch audit in <b>massgaps</b>                                                                                               |
| vacuum P0'                           | isostatic 2D, $\langle z \rangle = 4$                                   | postulated (v0.6)                            | BH = max-tension extreme; $d_I = 2$ predicted; $M_O$ reconstruction                                                                                                        |
| $d_{\text{eff}}$ protocol            | $d \ln V / d \ln r$ bracket                                             | measured (25 tests)                          | shortest below + diffusion above on lattices; shells fail (S11)                                                                                                            |
| D10 tension map                      | GR dip/overshoot/ $\rightarrow 3^+$                                     | GR computed; graph open                      | bare shortest-path rejected (inverted); D11 tail tension-dep. ( $\chi_1 1/r^2$ , $\chi_2 1/r$ ); $\kappa$ -profile closed (disk confirms)                                  |
| D12 reconstruction                   | $\forall M_O \in \mathcal{A}_{\text{macro}}$ same IR                    | open program                                 | access floor + non-circularity; P4 audit 1,2,3,5 pass, 4 partial; vantage axis filed; static depth negative                                                                |
| D13 causal order                     | D13.0 control measured; stages 1-4 open                                 | static control measured (S11)                | $V_G$ , $8r+4$ law; family (tri/hex/Del/Lloyd/Gab/kNN/medial +, gated-wall/rewired -); sweep: $N^* O(1)$ range, $f^* \rightarrow 0$ hypoth.; spectral $d_s$ band; twins C3 |
| $U$ -rule search                     | locality protection by $U$                                              | tournament open; single-move insufficient    | 6 rules stall w/ residual longs; ( $p, \text{longs}$ ) joint falsifier; $p$ -healing gameable                                                                              |
| $\kappa \rightarrow c_2$ map         | $c_2 = p(2p - 1)$                                                       | ansatz (open derivation)                     | power-law vs $1/r^2$ disagree cross-applied                                                                                                                                |
| evap. isometry $V_k$                 | qubit $V_t$ done (D2)                                                   | graph $V_k$ open (S3)                        | $V_t^\dagger V_t = I$ tested; graph derivation queued (D1)                                                                                                                 |
| Kerr $M_2$ , $g_{t\phi}$ , ISCO, QNM | area only                                                               | not derived (S3)                             | $A(M, a)$ assumed; quadrupole/QNM open; GW250114 QNM + GW241011                                                                                                            |
| echo $R$                             | $10^{-80}$ ampl., $10^{-160}$ energy                                    | dimensional estimate                         | $\delta_Q$ future wires<br>$\propto \omega^2$ lattice scattering, not leg S-matrix; $\sim 158$ orders margin                                                               |

Table 2: Gradient-shell  $p$  measurements:  $\beta$  recalibrated per  $N$  (log-linear over 6 points). All 80-graph production rows use exact distances; errors are per-graph std with SEM quoted in text.

| $N$   | shells $\times$ per-shell | $\beta$ | measured $p$                                                 | backend             |
|-------|---------------------------|---------|--------------------------------------------------------------|---------------------|
| 300   | $10 \times 30$            | 1.50    | $0.94 \pm 0.06$ (4 graphs)                                   | NetworkX exact      |
| 600   | $10 \times 60$            | 1.28    | $0.96 \pm 0.06$                                              | NetworkX exact      |
| 1020  | $10 \times 102$           | 1.24    | $0.913 \pm 0.049$ , SEM 0.0055, stacked<br>0.911 (80 graphs) | Floyd+LP exact      |
| 4000  | CSR-direct                | 0.99    | $0.9315 \pm 0.0032$ (80 graphs)                              | CPU + Sinkhorn OR   |
| 8000  | CSR-direct                | 0.87    | $0.9382 \pm 0.0030$ (80 graphs)                              | pod + Sinkhorn OR   |
| 16000 | CSR-direct                | 0.74    | $0.9137 \pm 0.0022$ (80 graphs)                              | GPU farm + torch OR |

Table 3: PPN status: only  $\gamma$  is claimed.  $\beta$  is coordinate-confused and not quoted.

| param                | GR     | this model                                                                 |
|----------------------|--------|----------------------------------------------------------------------------|
| $\gamma$             | 1      | 1 exactly (tortuosity $h = (1 + x/2)^2$ ; BV $c \rightarrow \gamma = 2c$ ) |
| $\beta$              | 1      | unresolved (not quoted)                                                    |
| $\alpha_{1,2}, \xi$  | 0      | untouched                                                                  |
| Nordtvedt            | 0      | untouched                                                                  |
| 2PN $c_{\text{tot}}$ | 4.8695 | 4.8693 at $p = 0.92$ (J0737 0.1 $\sigma$ )                                 |

Table 4: What exists today: no qualifying test yet. Only multi-detector,  $< 100 \text{ deg}^2$  gap events count.

| event     | masses                              | distance              | localisation / EM                                                                            | verdict                                                                                    |
|-----------|-------------------------------------|-----------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| GW190814  | $23.2 + 2.6$ ( $q \sim 0.11$ )      | $241^{+41}_{-45}$ Mpc | $23 \text{ deg}^2$ (3-det.); CFHT g<br>$22.8@1.7\text{d}$ 65.5%, GROWTH i<br>6 epochs to 98% | <b>tension:</b> $P(\text{detect}) \approx 0.68$ –<br>$0.88$ , $p \sim 0.12$ ; not excluded |
| GW230529  | $2.5\text{--}4.5 + 1.2\text{--}2.0$ | $\sim 197$ Mpc        | $24,200 \text{ deg}^2$ (single-det.); ZTF<br>7% to $g = 21.1$                                | not a test: 93% unsearched;<br>$m_g \sim 21.2$ at depth edge                               |
| S250206dm | mgBH candidate                      | —                     | no counterpart (per notes)                                                                   | unverified: check GCNs before citing                                                       |